Question			Answer	Marks	AO	Guidance
11	(a)		If geometric, then $r = \frac{3k-6}{5k-2}$	M1	2.4	Allow instead for $r = \frac{k+2}{3k-6}$ or $r^2 = \frac{k+2}{5k-2}$ in any form Soi
			Common ratio gives $\frac{3k-6}{5k-2} = \frac{k+2}{3k-6}$ So $9k^2 - 36k + 36 = 5k^2 + 8k - 4$	M1	2.1	Forms an equation in k which need not be simplified
			So $k^2 - 11k + 10 = 0$	A1	2.1	**AG** Rearranges to correct three term quadratic. At least one intermediate step must be shown.
						SC1 for showing $k = 1$ leads to $3, -3, 3$ $(r = -1)$ and that $k = 10$ leads to 48, 24, 12 $(r = \frac{1}{2})$ and demonstrating that both are geometric
				[3]		
11	(b)		So $k = 1, 10$	M1	1.1a	Solves the quadratic to give at least one root Soi
			When $k = 1$ the sum of 20 terms is $(3+(-3))+(3+(-3))+...+(3+(-3)) = 0$	M1 A1	3.1a 1.1	Evaluating the terms of the sequence when $k = 1$ cao
			Alternative for the last 2 marks $S_{20} = \frac{3(1-(-1)^{20})}{1-(-1)} = 0$	M1 A1		Using the formula for the sum of terms of a GP with $r = -1$ cao
				[3]		
11	(c)		When $k = 10$ the sequence is 48, 24, 12…			
			So $a = 48, r = \frac{1}{2}$	B1	3.1a	Identifies the first term and common ratio soi
			$S_\infty = \frac{48}{1-\frac{1}{2}} = 96$	B1	1.1	cao
				[2]		